

Inducing Neural Collapse via Anticlasses and One-Cold Cross-Entropy Loss

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Abstract—While softmax cross-entropy (CE) loss is the standard objective for supervised classification, it primarily focuses on the ground-truth classes, ignoring the relationships between the nontarget, complementary classes. This leaves valuable information unexploited during optimization. In this work, we propose a novel loss function, one-cold CE (OCCE) loss, which addresses this limitation by structuring the activations of these complementary classes. Specifically, for each class, we define an *anticlass*, which consists of everything that is not part of the target class—this includes all complementary classes as well as out-of-distribution (OOD) samples, noise, or in general any instance that does not belong to the true class. By setting a uniform one-cold encoded distribution over the complementary classes as a target for each anticlass, we encourage the model to equally distribute activations across all nontarget classes. This approach promotes a symmetric geometric structure of classes in the final feature space, increases the degree of neural collapse (NC) during training, addresses the independence deficit problem of neural networks, and improves generalization. Our extensive evaluation shows that incorporating OCCE loss in the optimization objective consistently enhances performance across multiple settings, including classification, open-set recognition, and OOD detection.

Index Terms—Classification, neural collapse (NC), neural networks, open set recognition, representation learning.

I. INTRODUCTION

DEEP learning has seen amazing progress over the past decade across multiple domains such as vision, speech, and text [1]. The optimization criterion universally used for supervised classification tasks is the cross-entropy (CE) loss, which aims at minimizing the discrepancy between the predicted probability distribution across classes and the ground-truth probability distribution, typically represented as a one-hot encoding of the true label [2]. Minimizing the CE loss maximizes the likelihood of correctly identifying disjoint classes under a given set of parameters of a statistical learning model, such as a neural network, making it an intuitive choice, with strong empirical results over the years.

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Despite its theoretical and practical appeal, CE loss is not without its limitations. Neural networks optimized with CE loss are known to result in overconfident predictions in incorrect classes [3], [4]. This problem becomes more evident when dealing with imbalanced data [5], or in open set classification scenarios, where a classification model must be able to also recognize unknown (i.e., not seen in training) classes at inference time [6], [7]. Current neural networks struggle to recognize out-of-distribution (OOD) samples [8] and often misclassify these instances to one of the known classes with high confidence, which calls into question their trustworthiness. In recent work, Feng et al. [9] also identified an independence deficit in neural networks, where the prediction confidence of some classes is redundantly determined by others through simple linear relationships, even when the classes are semantically unrelated. This entanglement can lead to overfitting and poor generalization, especially under domain shifts, as the network neglects to learn distinct class representations.

Despite these limitations, neural networks can generalize remarkably well [10], and understanding the underlying reasons has been the focus of extensive research [11], [12]. In this effort, recent work has uncovered the intriguing empirical phenomenon of neural collapse (NC), which arises during the terminal stage of deep neural network training [13], [14]. In NC, class features collapse to their respective means, class means become maximally separated in a simplex equiangular tight frame (ETF), and the classifier aligns with these class means, leading to a highly structured decision boundary. NC has been linked to improved robustness and benign overfitting [15], [16], with connections to the information bottleneck theory [17], which suggests that networks focus on essential information while discarding unnecessary variability.

While independence deficit and NC are distinct and even contradicting behaviors—one reflecting a redundancy in class independence and the other signifying maximally separated and independent class representations—CE loss does not provide a mechanism to explicitly control these phenomena. Instead, their emergence during training is often incidental, with networks passively exhibiting one behavior or the other, depending on the dataset size, class distribution, task difficulty, and model complexity rather than actively optimizing for either. The main novelty of this work is the rethinking of classification supervision through the introduction of the concept of an *anticlass*, which consists of the complementary set of a class, encompassing all that the class is not. With anticlasses, we can explicitly control the relationships between the complementary classes, and therefore the structure of the

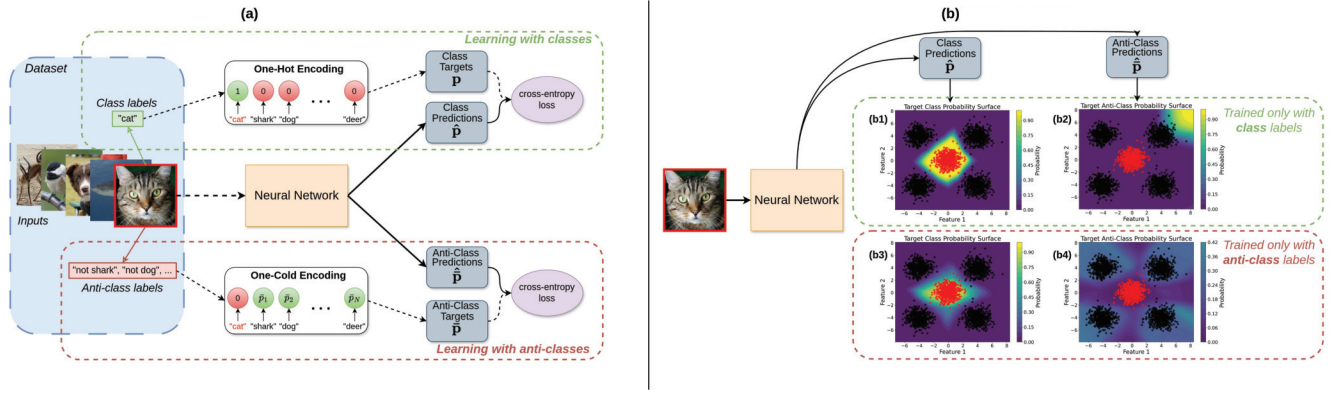


Fig. 1. (a) Training phase: standard classification uses one-hot encoding as the target distribution for classes. Our framework extends this approach by allowing the direct specification of target distributions for both classes and anticlasses—the latter indicating the degree to which each class does not match the ground truth. These options can be used independently or in combination. (b) Inference phase: the learned probability surfaces of a two-layer MLP for five Gaussian classes are illustrated over the initial input space. We compute both class predictions using the softmax of logits and anticlass predictions using the softmax of negated logits. (b1) One-hot CE produces sharp class boundaries, which can lead to overconfident predictions. (b2) Furthermore, the logits for the complementary classes are not equal, resulting in skewed anticlass distributions. (b3) In contrast, OCCE results in calibrated class decision boundaries. (b4) It also enforces a uniform anticlass distribution across all complementary classes.

classes in the final feature manifold. Fig. 1(a) illustrates the proposed framework, which enables both class and anticlass learning, while Fig. 1(b) highlights the nontrivial interactions that arise between these two approaches when used in isolation, affecting the output probability surface for a given class.

Inspired by the geometric properties of the Simplex ETF, we adopt a uniform one-cold encoding, where each complementary class is labeled with 1 and the true class is labeled with 0. This encoding reflects that all complementary classes of a target class belong equally to its anticlass. This label transformation ensures that the nonzero targets in the complementary classes encourage each class to maintain a symmetric relationship with all other classes, similar to the symmetry in the Simplex ETF, where class centroids are equinorm and equiangular. This approach promotes the network to develop distinct and orthogonal representations for each class, preventing reliance on linear correlations with other classes. As a result, the increased angular separation between class vectors enlarges interclass margins and lowers generalization error [18], [19]. Furthermore, converging to a Simplex ETF configuration simplifies the class structure, enabling the classifier to operate effectively as a nearest-centroid classifier—a behavior that has been theoretically linked to improved generalization in the NC regime [13].

To summarize the contributions of this work, we first introduce the concept of an anticlass, providing a novel approach to modeling classification problems by directly optimizing the relationships between complementary classes. We then propose one-cold CE (OCCE) loss, which enables the network to learn and predict anticlass distributions, allowing it to control the geometric structure of classes. Our comprehensive analysis and empirical evaluation demonstrate that incorporating anticlasses provides a nontrivial and complementary training signal that enhances optimization and improves performance in both closed-set and open-set classification tasks. Finally, we analyze the impact of uniform OCCE on NC, independence deficit, and the generalization capabilities of neural networks.

The remainder of this article is structured as follows. In Section II, we briefly discuss related work and compare it to the proposed method. Section III presents the derivation of the proposed method, followed by an extended experimental evaluation in Section V, which includes closed-set classification, open-set recognition, and OOD experiments. Finally, conclusions are drawn in Section VII.

II. RELATED WORK

Controlling the structure of class representations in the feature space and understanding how class geometry impacts model performance have been extensively studied in the literature [20]. In this section, we discuss the most relevant works and how they compare and differentiate from our proposed approach.

A. Regularizing Complementary Classes

Regularizing complementary classes was first explored in [21], where label smoothing was introduced to reduce neural network overconfidence. This technique redistributes a small fraction ϵ of the target probability from the true class to the complementary classes, producing the smoothed target vector

$$\mathbf{y}'_i = (1 - \epsilon)\mathbf{y}_i + \frac{\epsilon}{N} \quad (1)$$

where N is the number of classes, $\mathbf{y}$ represents the original target vector, and $\mathbf{y}'$ is the smoothed target vector. However, label smoothing introduces bias by lowering the target probability assigned to the true class. In contrast, our proposed loss maintains asymptotic consistency with one-hot encoding.

Complement objective training [22] alternates between optimizing the CE loss and a complement objective, which maximizes Shannon's complement entropy, defined as

$$C(\hat{\mathbf{y}}) = - \sum_{j=0, j \neq c}^N \frac{\hat{y}_j}{\hat{y}_c} \log \frac{\hat{y}_j}{\hat{y}_c} \quad (2)$$

where c denotes the target class, $\hat{\mathbf{y}}$ is the predicted vector representing probabilities over the classes, and N is the number of classes. This objective surpasses the predicted probabilities of complementary classes with high confidence. The dual-optimizer approach requires two updates per batch, increasing training time by a factor of 1.6. Later, Kim et al. [23] unified these objectives into a single loss, termed complementary CE (CCE) loss. In contrast, OCCE sets not-zero targets to complementary classes and can, therefore, function as a stand-alone objective. It also provides the flexibility to incorporate any target distribution for the complementary classes. For example, soft anticlass distributions can be used, inspired by knowledge distillation techniques, such as [24] and [25].

B. Adversarial RPL

Reciprocal point learning (RPL) models the “otherness” of a class by positioning a reciprocal point $\mathbf{P}_k \in \mathbb{R}^d$ for each known class k in the d -dimensional latent space [26]. In this framework, a sample’s latent vector $\mathbf{h} \in \mathbb{R}^d$ is compared to each $\mathbf{P}_k$ for all classes via a distance measure $d_e(\mathbf{h}, \mathbf{P}_k)$, and classification is performed using a softmax over these distances. Adversarial RPL (ARPL) extends this idea by introducing the following adversarial margin constraint:

$$L_o(\mathbf{x}) = \max(d_e(\mathbf{h}, \mathbf{P}_k) - R, 0) \quad (3)$$

where $\mathbf{x}$ is the input sample, $\mathbf{h}$ is its latent representation, and R is a predefined margin threshold that restricts the open space and reduces the risk of misclassifying unknown samples [27]. ARPL + CS incorporates augmented adversarial confusing samples from the confrontation between the known data and reciprocal points. The main difference between ARPL and our proposed method is analogous to the contrast between prototype-based and likelihood-based classification approaches.

C. Simplex ETF Classifiers

NC suggests that the final classifier layer converges to a Simplex ETF, where the class-means converge to having equal length, forming equal-sized angles between any given pair, and being the maximally pairwise-distanced configuration constrained to the previous two properties [13]. The standard K -Simplex ETF is a collection of points in $\mathbb{R}^K$ specified by the columns of a matrix $\mathbf{M}$, defined as

$$\mathbf{M} = \sqrt{\frac{K}{K-1}} \left(\mathbf{I} - \frac{1}{K} \mathbf{1}_K \mathbf{1}_K^\top \right) \quad (4)$$

where K is the number of points in the ETF, $\mathbf{I} \in \mathbb{R}^{K \times K}$ is the identity matrix, and $\mathbf{1}_K \in \mathbb{R}^K$ is the vector of ones. Zhu et al. [14] and Pernici et al. [28] fix the last layer classifier to be a Simplex ETF, demonstrating time and memory gains without performance loss. Furthermore, Simplex ETF structure has been used to mitigate minority collapse in imbalanced training [29] and few-shot incremental learning [30]. OCCE diversifies from these approaches because it induces the Simplex ETF structure under a controllable and end-to-end learnable paradigm, rather than predefining the structure of the classifier. Furthermore, our approach allows setting any desirable

anticlass target distribution to control different tradeoffs. This enables controlling the tradeoff between enforcing NC versus accounting for interclass diversity, which has been proven beneficial in works like [31].

III. PROPOSED METHOD

A. Ordinary Supervised Classification

In a standard supervised learning scenario, we deal with a set of n labeled instances $\mathcal{D} = \{(\mathbf{x}_0, l_0), \dots, (\mathbf{x}_n, l_n)\}$ over an initial feature space $\mathcal{X} \subseteq \mathbb{R}^d$ and a discrete label space $L = \{1, \dots, N\}$, where d is the dimensionality of $\mathcal{X}$ and N is the number of classes. An instance $\mathbf{x}$ that belongs to class k is sampled from the subspace $\mathcal{C}_k$. Given a model with parameter θ , the objective of an ordinary classification task is to find the mapping $f: \mathcal{X} \rightarrow L$, where $f(\mathbf{x}; \theta) = l$, correctly mapping each input $\mathbf{x}$ to its corresponding label l . Standard classification training minimizes the CE loss for each instance, comparing the model’s predicted probability distribution $\hat{\mathbf{y}}$ with the one-hot encoded label $\mathbf{y}$. The CE loss is defined as

$$L_{CE} = -\mathbf{y}^\top \log(\hat{\mathbf{y}}), \quad \text{where } \hat{y}_i = \frac{\exp(z_i)}{\sum_{j=1}^N \exp(z_j)}. \quad (5)$$

In (5), only the log probability of the ground-truth labels is explicitly minimized, while the probabilities of the complementary classes are indirectly influenced through the softmax normalization. This approach overlooks the relationships between the complementary classes. Such a limitation can be problematic, particularly for handling anomalous samples like adversarial examples or OOD inputs. Ideally, for these cases, a model should produce a close to uniform probability distribution across all classes, indicating maximum uncertainty. However, achieving this uniform distribution requires equal activations $\mathbf{z}$ for all classes, which is not explicitly encouraged by CE loss. Furthermore, it does not enforce any specific geometric structure of the classes in the feature space, leading to undesirable phenomena such as minority collapse [32], where features of minority classes collapse together, or independence deficit [9], where spurious correlations between classes emerge.

B. Anticlass Formulation

To address these challenges, we shift our focus to anticlasses. Let us consider a predefined set of N known classes. Each class k can be represented as a subspace in a d -dimensional space, denoted by $\mathcal{C}_k \subseteq \mathbb{R}^d$. For every class k , we can define the complementary subspace to $\mathcal{C}_k$, which we denote as $\bar{\mathcal{C}}_k = \mathbb{R}^d \setminus \mathcal{C}_k$ and refer to as the anticlass subspace of k . The anticlass subspace $\bar{\mathcal{C}}_k$ contains every instance that does not belong to class k , including instances from other known or unknown classes, OOD instances, random noise, or adversarial examples. Intuitively, directly modeling the anticlass subspace $\bar{\mathcal{C}}_k$ seems a much harder problem than modeling $\mathcal{C}_k$, as the latter is clearly more restricted. Moreover, in mutually exclusive classes, an instance that belongs to $\mathcal{C}_k$ inherently belongs to the anticlass of all other known classes $\bar{\mathcal{C}}_{j \neq k}$. Consequently, in multiclass classification, every vector $\mathbf{p} \in \mathbb{R}^d$ belongs to at least $N - 1$ anticlasses, and potentially to all N of them, if the

vector does not represent an instance of a known class. We can encode the anticlass labels using a one-cold encoded label $\bar{\mathbf{I}}$, where the ground-truth class is labeled with 0 (hence, the term cold), and every complement class is labeled with 1. The vector $\bar{\mathbf{I}}$ can be defined as

$$\bar{I}_i = \begin{cases} 0, & \text{if } i = \text{ground-truth class} \\ 1, & \text{if } i = \text{complementary class.} \end{cases} \quad (6)$$

C. Deriving the OCCE Loss

Since one-cold encoded labels are not valid probability distributions, standard CE loss cannot be directly applied. To control the relationships between complementary classes, we introduce competition between anticlasses by applying a softmax function. While in one-hot encoding, this results in a winner-takes-all scenario [2], with anticlasses, the target probability distribution can reflect how much an instance belongs to the anticlass corresponding to each complementary class. In this work, we adopt a uniform one-cold encoding, where a sample fully belongs to all anticlasses except its true class, implying that all anticlasses are treated equally as winners. The target anticlass probability distribution $\bar{\mathbf{y}}$ can be simply defined for each instance as

$$\bar{y}_i = \frac{\bar{I}_i}{\sum_{j=1}^N \bar{I}_j} = \begin{cases} 0, & \text{if } i = \text{ground-truth class} \\ \frac{1}{N-1}, & \text{if } i = \text{complementary class.} \end{cases} \quad (7)$$

Let now $\bar{\mathbf{z}}$ be the output activation vector of our network. For a network optimized to learn anticlasses, higher activations correspond to a higher probability of not belonging to the corresponding class. We can now define the OCCE loss as

$$L_{\text{OCCE}} = -\bar{\mathbf{y}}^T \log(\hat{\mathbf{y}}), \quad \text{where } \hat{y}_i = \frac{\exp(\bar{z}_i)}{\sum_{j=1}^N \exp(\bar{z}_j)}. \quad (8)$$

Finally, to obtain class activations $\mathbf{z}$, we negate the anticlass activations $\mathbf{z} = -\bar{\mathbf{z}}$ before softmax.

D. Convergence Analysis

Let us consider the general CE loss, given by

$$L = -\sum_{i=1}^N y_i \log(\hat{y}_i) \quad (9)$$

where $\mathbf{y}$ is any target distribution and $\hat{y}_i$ is the predicted probability for class i , defined by a softmax function over the class activations $\mathbf{z}$. The partial derivative of the loss with respect to $\hat{y}_i$ is

$$\frac{\partial L}{\partial \hat{y}_i} = -\frac{y_i}{\hat{y}_i} \quad (10)$$

while the softmax derivative has two cases

$$\frac{\partial \hat{y}_i}{\partial z_j} = \begin{cases} \hat{y}_i(1 - \hat{y}_i), & \text{if } i = j \\ -\hat{y}_i \hat{y}_j, & \text{if } i \neq j. \end{cases} \quad (11)$$

Using the chain rule and substituting, we get

$$\frac{\partial L}{\partial z_j} = \sum_{i=1}^N \frac{\partial L}{\partial \hat{y}_i} \cdot \frac{\partial \hat{y}_i}{\partial z_j}$$

$$\begin{aligned} &= \sum_{i \neq j} \underbrace{-\frac{y_i}{\hat{y}_i} \cdot (-\hat{y}_i \hat{y}_j)}_{i \neq j} + \underbrace{\left(-\frac{y_j}{\hat{y}_j} \cdot \hat{y}_j(1 - \hat{y}_j)\right)}_{i=j} \\ &= \sum_{i \neq j} y_i \hat{y}_j + y_j(\hat{y}_j - 1). \end{aligned} \quad (12)$$

Substituting the target distributions for the standard one-hot encoding $\mathbf{y}$ and our proposed one-cold encoding $\bar{\mathbf{y}}$ defined for a correct class c as

$$y_i = \begin{cases} 1, & \text{if } i = c \\ 0, & \text{if } i \neq c \end{cases} \quad \text{and} \quad \bar{y}_i = \begin{cases} 0, & \text{if } i = c \\ \frac{1}{N-1}, & \text{if } i \neq c \end{cases}$$

we get the derivative of the standard CE L_{CE} as

$$\frac{\partial L_{\text{CE}}}{\partial z_i} = \begin{cases} \hat{y}_i - 1, & \text{if } i = c \\ \hat{y}_i, & \text{if } i \neq c \end{cases} \quad (13)$$

and the derivative of the OCCE L_{OCCE} as

$$\frac{\partial L_{\text{OCCE}}}{\partial \bar{z}_i} = \begin{cases} \hat{\bar{y}}_i, & \text{if } i = c \\ \hat{\bar{y}}_i - \frac{1}{N-1}, & \text{if } i \neq c. \end{cases} \quad (14)$$

For the asymptotic behavior of CE, as $\nabla_{\mathbf{z}} L_{\text{CE}} \rightarrow \mathbf{0}$, under no other constraints, we obtain

$$\hat{y}_c = \frac{\exp(z_c)}{\sum_j \exp(z_j)} \rightarrow 1 \implies z_c \rightarrow +\infty, \quad \text{and} \quad (15)$$

$$\hat{y}_k = \frac{\exp(z_k)}{\sum_j \exp(z_j)} \rightarrow 0 \implies z_k \rightarrow -\infty, \quad \text{for } k \neq c. \quad (16)$$

In the case of OCCE, when $\nabla_{\bar{\mathbf{z}}} L_{\text{OCCE}} \rightarrow \mathbf{0}$ and under no other constraints, we obtain

$$\begin{aligned} \hat{\bar{y}}_c &= \frac{\exp(-z_c)}{\sum_{j=1}^N \exp(-z_j)} \rightarrow 0 \\ &\implies \exp(-z_c) \rightarrow 0 \\ &\implies z_c \rightarrow +\infty, \quad \text{for } c \text{ and} \end{aligned} \quad (17)$$

$$\begin{aligned} \hat{\bar{y}}_k &= \frac{\exp(-z_k)}{\sum_{j=1}^N \exp(-z_j)} \rightarrow \frac{1}{N-1} \\ &\implies \exp(-z_k) \rightarrow \frac{1}{N-1} \sum_{j=1}^N \exp(-z_j) \\ &\implies \exp(-z_k) \rightarrow \frac{1}{N-1} \sum_{j=1, j \neq c}^N \exp(-z_j) \\ &\implies z_k \rightarrow -\log\left(\frac{\sum_{j \neq c} \exp(-z_j)}{N-1}\right) \\ &\implies z_k \rightarrow z_{\text{const}}, \quad \text{for any } k \neq c. \end{aligned} \quad (18)$$

Equation (18) suggests that all complementary classes k converge to the same value, i.e., $z_{k_1} \rightarrow z_{k_2}$ for any $k_1, k_2 \neq c$. This condition is essential for convergence to Simplex ETF, which CE can also theoretically achieve [14]. However, in practice, the exponential in the softmax function compresses the differences among lower valued logits in CE and as a result does not penalize them. In contrast, the OCCE loss amplifies these differences by applying the exponential to the negated logits and enforces their equality, clearly shown in Fig. 2.

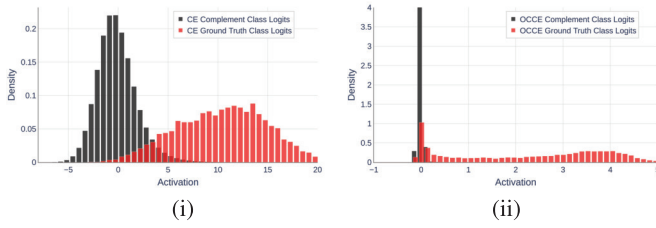


Fig. 2. CIFAR-100 logits distributions for target and complementary classes. OCCE forces the logits of all complementary classes to converge to the same value, in contrast to CE.

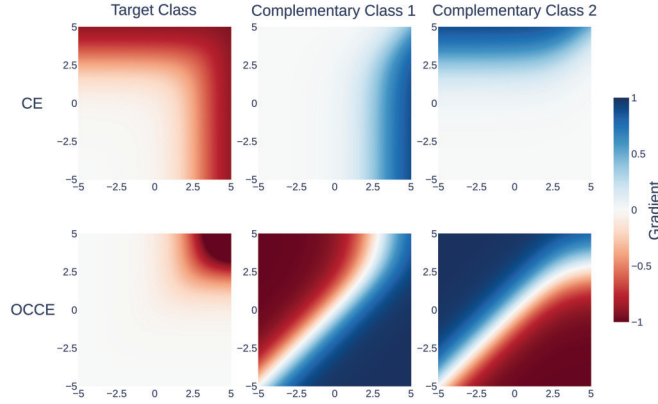


Fig. 3. Given a three-class classification scenario, we fix the ground-truth class activation $z_0 = 3$ and plot activations for complement classes z_1 and z_2 ranging from -5 to 5 . Each grid shows the gradients for each class at any (z_1, z_2) point.

E. Gradient Behavior

In Fig. 3, we analyze the gradient dynamics under the following setup: a three-class scenario where the activation for the correct class is fixed at $z_0 = 3$ and the activations for the two complementary classes, z_1 and z_2 , vary within the range $[-5, 5]$. Under CE, the correct class is encouraged to have higher activation than the complementary classes, which are independently driven toward lower values. Large gradients appear when either complementary class has a competitive activation, without considering their relationship. Consequently, for a classification task with N classes, gradients have a small magnitude as long as $\max_{j \neq c} z_j \ll z_c$. In contrast, under OCCE, if the activations of the complementary classes have different values, their gradients work to equalize them. As a result, high magnitude gradients are observed in most regions except where $\forall j \neq c: z_j = z_{\text{const}} \ll z_c$, which is a much more restricted area than that of CE.

IV. ARCHITECTURE DESIGNS WITH ANTICLASSES

A. Stand-Alone OCCE Architecture

OCCE can be used to train a neural network for classification on its own. To perform classification, we negate the activations $\bar{\mathbf{z}}$ to obtain class activations $\mathbf{z} = -\bar{\mathbf{z}}$, where higher activations to more probable classes. By passing $\mathbf{z}$ through softmax, we obtain the predicted class probability distribution $\hat{\mathbf{y}}$. Empirically, while this architecture is competitive in easier datasets, we observe convergence instabilities on

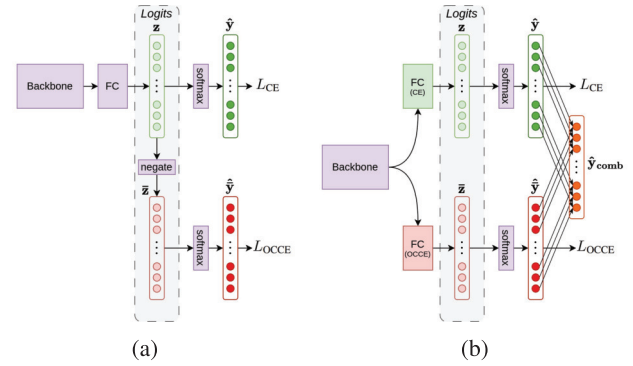


Fig. 4. Proposed architectures that efficiently combine class and anticlass learning in shared or decoupled layers. (a) Unified architecture. (b) Dual architecture.

more complex datasets with a larger number of classes, due to the increased challenge of aligning the activations of all complementary classes, and the decreased numerical training signal of each complementary class, as the number of classes gets larger. To overcome this limitation, we propose two architectures that can combine the benefits of OCCE with CE, ensuring a strong training signal to both target and nontarget classes. As we experimentally demonstrate in Section V-B, this approach outperforms all other methods in more challenging datasets (e.g., open set recognition (OSR) experiments).

B. Unified CE + OCCE Objective

The most straightforward way to use the OCCE objective as an added term to any CE-based loss is to apply it on the negated class activations ($\bar{\mathbf{z}} = -\mathbf{z}$), while applying the standard CE loss on $\mathbf{z}$, as shown in Fig. 4(a). The unified objective keeps a high signal for the target class through the CE term, while also aligning the activations of the complementary classes through uniform OCCE. Finally, a scalar γ controls the contribution of OCCE in the unified loss

$$L = L_{CE} + \gamma \cdot L_{OCCE}. \quad (19)$$

C. Dual-Head Architecture

An interesting observation from our experiments is that the earlier layers of models trained with CE and stand-alone OCCE learn highly similar representations. However, beyond a certain point, the representations diversify significantly. This can be seen using the centered kernel alignment (CKA) similarity matrix [33], which measures the similarity between the layerwise representations of two networks, as shown in Fig. 5. In Fig. 5(a), networks with different initializations trained with CE converge to similar final-layer representations, whereas in Fig. 5(b), CE and OCCE produce highly diverse representations. To gain deeper insights into this behavior, we experiment with decoupled projection heads for CE and OCCE, where each head has distinct initial random projection weights, while the shared backbone is optimized by gradients from both losses [see Fig. 4(b)]. This setup enables a direct comparison of linear probing under each loss within shared representations.

TABLE I
BASE TRAINING TEST ERRORS USING DIFFERENT LOSS FUNCTIONS. RESULTS ARE PRESENTED AS BASELINE
(WITHOUT OCCE, $\gamma = 0$) AND WITH OCCE ($\gamma = 1$)

Test Errors ($\downarrow$)		ResNet18v2		MobileNetv2		DenseNet121	
Dataset	Loss	Baseline	w/ OCCE	Baseline	w/ OCCE	Baseline	w/ OCCE
CIFAR-100	CE	24.98 $\pm$ 0.2	23.92 $\pm$ 0.2	30.03 $\pm$ 0.3	28.83 $\pm$ 0.2	25.06 $\pm$ 0.2	24.19 $\pm$ 0.1
	LSCE	23.26 $\pm$ 0.1	22.97 $\pm$ 0.1	29.26 $\pm$ 0.1	29.14 $\pm$ 0.3	23.59 $\pm$ 0.2	23.43 $\pm$ 0.1
	CE+NL	25.10 $\pm$ 0.1	23.90 $\pm$ 0.3	29.95 $\pm$ 0.2	28.89 $\pm$ 0.3	25.48 $\pm$ 0.8	24.71 $\pm$ 0.4
	COT	24.52 $\pm$ 0.1	23.21 $\pm$ 0.2	29.57 $\pm$ 0.1	28.52 $\pm$ 0.3	23.70 $\pm$ 0.7	22.27 $\pm$ 0.2
	CCE	25.36 $\pm$ 0.2	23.57 $\pm$ 0.2	29.81 $\pm$ 0.2	28.72 $\pm$ 0.3	25.29 $\pm$ 0.3	23.80 $\pm$ 0.3
TinyImageNet	CE	36.80 $\pm$ 0.2	35.61 $\pm$ 0.1	39.69 $\pm$ 0.4	38.79 $\pm$ 0.2	38.71 $\pm$ 0.6	36.49 $\pm$ 0.5
	LSCE	36.98 $\pm$ 0.1	36.01 $\pm$ 0.1	39.10 $\pm$ 0.1	39.08 $\pm$ 0.1	38.06 $\pm$ 0.5	37.52 $\pm$ 0.6
	CE+NL	36.40 $\pm$ 0.1	35.28 $\pm$ 0.1	39.16 $\pm$ 0.1	38.41 $\pm$ 0.2	38.75 $\pm$ 1.2	38.66 $\pm$ 1.0
	COT	35.40 $\pm$ 0.1	34.74 $\pm$ 0.1	40.14 $\pm$ 0.2	39.21 $\pm$ 0.2	37.60 $\pm$ 0.2	35.11 $\pm$ 0.1
	CCE	36.88 $\pm$ 0.1	35.45 $\pm$ 0.1	39.19 $\pm$ 0.3	38.81 $\pm$ 0.1	39.22 $\pm$ 0.1	37.47 $\pm$ 0.1

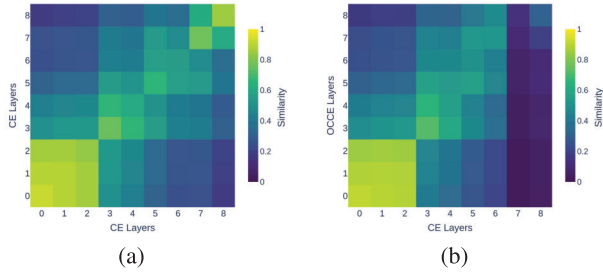


Fig. 5. CKA matrix between two networks trained with CE using different seeds (left) and two networks trained with CE and OCCE, respectively (right). (a) CE versus CE. (b) CE versus OCCE.

V. EXPERIMENTAL EVALUATION

In this section, we evaluate the proposed method across closed-set classification, open-set recognition, and OOD detection to explore the performance of OCCE in various settings. Then, in Section VI, we conduct a sensitivity analysis to examine its impact on NC, independence deficit, and generalization, thereby providing a detailed view of OCCE's empirical behavior that supports the analyses in Sections III-D and III-E.

A. Closed-Set Classification

1) *Base Training*: For our base training experiments, we train the following architectures: a ResNet18v2 [34], a MobileNetv2 [35], and a DenseNet121 [36] on CIFAR-100 [37] and TinyImageNet [38] datasets, containing 100 and 200 classes, respectively. We follow a standard hyperparameter configuration across all experiments from [39], which involves an SGD optimizer with a momentum of 0.9, a batch size of 128, a weight decay of 0.0001, and a total of 150 epochs for CIFAR-100 and 90 epochs for TinyImageNet. The learning rate starts from the base value of 0.1 and is divided by 0.1 at 50 and 100 epochs for CIFAR-100, and at 30, 60, and 80 epochs for TinyImageNet. For augmentation, we follow normalization, random horizontal flipping, and random cropping in the train set, while only normalizing and center cropping the images in the test set.

We make use of the unified objectives architecture [Fig. 4(a)], considering different baseline losses and adding the OCCE objective on top of them. Specifically, we consider:

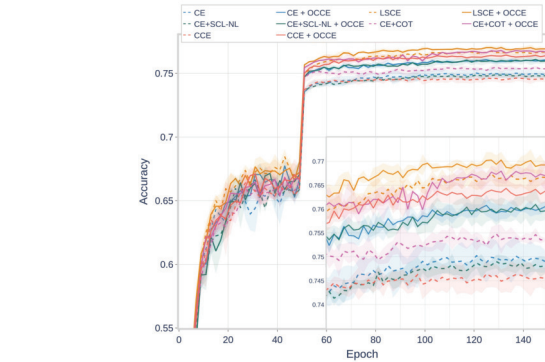


Fig. 6. Validation accuracy curves of CIFAR-100 with ResNet18v2.

TABLE II
TRANSFER LEARNING TEST ERRORS

Model	Dataset	Test Errors ($\downarrow$)	
		CE ($\gamma = 0$)	CE w/ OCCE ($\gamma = 1$)
WideResNet50	STL-10	1.87 $\pm$ 0.07	1.76 $\pm$ 0.08
	CIFAR-100	13.58 $\pm$ 0.19	13.35 $\pm$ 0.08
	TinyImageNet	17.33 $\pm$ 0.14	16.92 $\pm$ 0.24
	ImageNet	21.90 $\pm$ 0.22	21.15 $\pm$ 0.12
Swin-T	STL-10	2.27 $\pm$ 0.05	2.07 $\pm$ 0.07
	CIFAR-100	12.69 $\pm$ 0.07	12.41 $\pm$ 0.12
	TinyImageNet	14.45 $\pm$ 0.05	14.31 $\pm$ 0.02
	ImageNet	19.69 $\pm$ 0.02	19.33 $\pm$ 0.04

1) CE loss; 2) label smoothing with $\epsilon = 0.1$ (LSCE) [21]; 3) CE combined with negative log loss (CE + NL), a complementary label loss introduced for learning with noisy labels [40]; 4) complement objective training (COT) [22]; and 5) complement CE (CCE) [23] as our main competitors.

Table I presents the results of base training experiments, showing pairwise comparisons between different losses without and with the OCCE objective. It is evident that including OCCE leads to consistent improvements along all baselines and model combinations, suggesting that OCCE offers additional gains on top of the other regularization techniques, which is also confirmed from the validation accuracy curves in Fig. 6.

2) *Transfer Learning*: We finetune a WideResNet50 [41] and a Swin-T vision Transformer [42] on STL-10 [43], CIFAR-100, TinyImageNet, and ImageNet-1k [44]. We use

TABLE III
NER F -SCORE TEST ERRORS

F1 Errors ($\downarrow$)	CE ($\gamma = 0$)	CE w/ OCCE ($\gamma = 1$)	Comp. Overhead (%)
RoBERTa	4.38 $\pm$ 0.05	4.23 $\pm$ 0.07	+ 1.68 %
DistilBERT	6.34 $\pm$ 0.13	6.26 $\pm$ 0.20	+ 3.02 %
GPT-2	23.63 $\pm$ 0.41	23.26 $\pm$ 0.38	+ 1.07 %

the ImageNet-1k pretrained weights provided by PyTorch, remove the classification head, and initialize a new one. For hyperparameters, we use the configuration suggested by Kolesnikov et al. [45], which involves performing 10 000 updates on all medium-sized datasets, except the larger ImageNet dataset where 20 000 updates are performed. SGD optimizer is employed with 0.9 momentum and a base learning rate of 0.01. We decay the learning rate by a factor of 10 at 30%, 60%, and 90% of the training steps and use no weight decay and no dropout.

Table II presents the results of finetuning. Even though the pretrained weights were trained with standard CE, including OCCE in the finetuning process results in consistent improvements in accuracy across all settings.

To further evaluate the effectiveness of OCCE as a regularizer, we apply it to a natural language processing (NLP) downstream task, specifically to named entity recognition (NER). We consider the CoNLL-2003 dataset, which contains four entity and one nonentity classes [46], and three pretrained LLMs: RoBERTa [47], DistilBERT [48], and GPT-2 [49]. For fine-tuning the pretrained language models, we employ a standard hyperparameter configuration. This involves AdamW as an optimizer, a sweep of learning rates ($\text{lr} \in 10^{-4} \cdot [1, 2, 4]$), a batch size of 32, and a total of three epochs of fine-tuning with five different seeds per experiment. We evaluate the overall $F1$ -score to ensure fairness across entities. The results are shown in Table III, where OCCE consistently improves performance.

In NER, each token is classified individually, so OCCE adds an extra softmax per token during training, potentially resulting in computational overhead. In our experiments, it resulted in only a small overhead, a 2% average increase in training time (Table III). For longer sequences or more entities, limiting target anticlass distributions to competitive complementary entities may help mitigate additional overhead.

B. Open-Set Recognition

In OSR, we train the network on K known classes and test with samples from both known and U unknown classes. The typical method to enable the reject option for the unknown classes is through maximum over softmax probabilities (MSPs) [50]. In MSP, the goal is to assign lower confidence to unknown instances and higher to known ones, enabling unknown rejection by threshold.

1) *Configuration*: We select a configuration that enables comparison with previous benchmarks while extending them to more challenging scenarios. Following [27] and [51], we use the same splits and network architecture for MNIST and SVHN, where four classes are assigned as known and six as unknown. For TinyImageNet and CIFAR-100, we employ a

slightly larger architecture, specifically a ResNet18v2. In the case of TinyImageNet, we follow the standard split used in these prior works (20 known and 180 unknown classes). For CIFAR-100, we define a setup with 50 known and 50 unknown classes, to examine a higher number of known classes.

2) *Competitors*: We compare against CE, binary CE (BCE), generalized convolutional prototype learning (GCPL) [51], RPL [26], ARPL [27], and ARPL with confusing samples (ARPL + CS). CS involves simultaneously training a GAN to generate adversarial samples, for which the classification network increases uncertainty by maximizing the entropy of the predictions, resulting in an average training time of approximately four times longer in our experiments.

For our proposed framework, we explore different instantiations to examine their effectiveness. We train models with stand-alone OCCE, Unified CE + OCCE [see Fig. 4(a)], and our dual-head architecture, for which we evaluate performance using predictions from either the CE head, the OCCE head, or their combined outputs [see Fig. 4(b)]. To ensure a fair comparison with ARPL + CS, we incorporate the same augmented confusing samples into our framework. Specifically, we assign a uniform target anticlass distribution across all classes for these samples since they do not belong to any class. In our methods, we select the γ value for each dataset based on hold-out validation over the values $\gamma \in [0.2, 1, 5]$.

3) *Evaluation*: Since the rarity of unknown samples cannot be determined beforehand, OSR evaluation approaches requiring arbitrary thresholds are impractical. We use the following metrics instead.

- 1) *Correct Classification Rate (CCR)*: It measures the closed set accuracy, without considering the unknown samples.
- 2) *Area Under the Receiver Operating Characteristic (AUROC)*: A threshold-independent metric that measures the probability that a known sample is assigned higher MSP than an unknown.
- 3) *Open Set Classification Rate (OSCR)*: Introduced in [27] as a metric that considers both closed and open set performance, measuring correctly classified known samples against incorrectly classified unknown samples for various MSP thresholds.

All experiments were run on five different splits, with each split having a specified number of known and unknown classes, and three seeds were used per split.

4) *Results*: Table IV presents the results of our OSR experiments. The following observations can be made for the configurations without confusing samples. Stand-alone OCCE achieves the best performance on the simple MNIST dataset; however, as task complexity and the number of known classes increase, it struggles to remain competitive. This issue is effectively addressed by both unified and dual-head architectures, which outperform all competitors across all other datasets. Of particular interest is the performance gap between the CE and OCCE heads in the dual-head architecture, where both share the same feature space. Notably, the linear probing of OCCE consistently achieves a higher OSCR than CE across all cases. This demonstrates that OCCE produces better-calibrated decision boundaries compared to CE.

TABLE IV

OSCR AND AUROC/CCR RESULTS, AVERAGED OVER 15 RANDOMIZED TRIALS (FIVE SPLITS AND THREE SEEDS). FOR EACH EXPERIMENT, WE SELECT THE EPOCH THAT MAXIMIZES OSCR AND REPORT THE CORRESPONDING AUROC AND CCR VALUES. THE HIGHEST OSCR VALUE FOR EACH DATASET IS SHOWN IN BOLD, WHILE THE BEST OSCR AMONG METHODS WITHOUT AUGMENTATION IS UNDERScoreD

Method	MNIST (K=6/U=4)		SVHN (K=6/U=4)		CIFAR-100 (K=50/U=50)		TinyImageNet (K=20/U=180)	
	OSCR ($\uparrow$)	AUROC / CCR ($\uparrow$)	OSCR ($\uparrow$)	AUROC / CCR ($\uparrow$)	OSCR ($\uparrow$)	AUROC / CCR ($\uparrow$)	OSCR ($\uparrow$)	AUROC / CCR ($\uparrow$)
CE	99.21	99.38 / 99.70	92.60	94.42 / 96.50	64.49	73.95 / 78.05	67.05	76.62 / 79.72
BCE	99.38	99.55 / 99.75	93.11	94.81 / 96.85	64.93	74.73 / 78.69	69.47	78.91 / 80.88
GCP	98.91	99.06 / 99.75	92.41	94.04 / 96.85	27.16	57.50 / 34.24	66.46	76.86 / 78.84
RPL	99.36	99.50 / 99.75	92.71	94.38 / 96.78	50.51	63.11 / 71.31	65.22	75.81 / 78.43
ARPL	99.33	99.55 / 99.71	93.57	95.80 / 96.53	63.66	75.05 / 76.97	68.97	79.67 / 80.25
OCCE	99.44	99.59 / 99.78	93.27	95.05 / 99.80	65.56	75.23 / 78.56	69.87	79.60 / 80.52
Unified CE+OCCE	99.36	99.52 / 99.76	93.83	95.57 / 96.97	<u>66.62</u>	75.82 / 79.77	70.16	79.58 / 81.25
Dual (CE head)	99.36	99.52 / 99.75	93.62	95.24 / 96.99	65.98	74.99 / 79.34	69.08	78.07 / 80.79
Dual (OCCE head)	99.40	99.59 / 99.75	<u>93.89</u>	95.72 / 96.92	66.12	75.90 / 78.87	<u>70.54</u>	80.49 / 81.10
Dual (Comb. head)	99.38	99.55 / 99.75	93.83	95.55 / 96.98	66.40	75.76 / 79.27	<u>70.23</u>	79.62 / 81.20
ARPL +CS	99.36	99.57 / 99.72	93.67	96.00 / 96.49	65.97	76.29 / 78.99	70.27	80.68 / 80.92
Unified CE+OCCE +CS	99.33	99.48 / 99.74	94.21	95.82 / 97.20	67.93	76.64 / 80.69	71.05	80.07 / 81.65

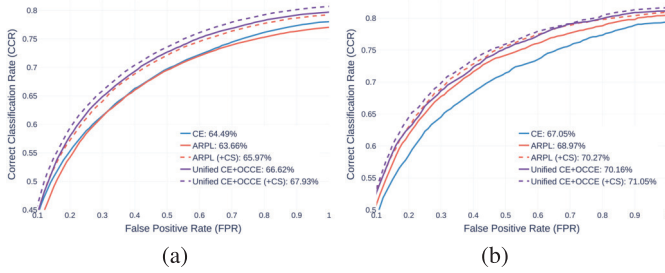


Fig. 7. OSCR ROC curves for ARPL and Unified CE + OCCE, without confusing samples (solid lines) and with confusing samples (dashed lines). CE is also shown for reference. (a) CIFAR-100 ($K = 50/U = 50$). (b) TinyImageNet ($K = 20/U = 180$).

Regarding the confusing samples, as shown in the last two rows of Table IV, this augmentation technique results in significant improvements across all datasets except MNIST, where no notable gains are observed. This can be attributed to the dataset's inherent simplicity. Unified CE + OCCE (+CS) outperforms ARPL (+CS), with a more significant gap in the more challenging CIFAR-100 and TinyImageNet datasets, where a greater number of known classes is used. In Fig. 7, we present the OSCR ROC curves for CE, ARPL, and Unified CE + OCCE (with and without CS). The results show that ARPL struggles to remain competitive and even underperforms compared to the CE baseline in CIFAR-100 where the number of known classes is higher. In contrast, the proposed Unified CE + OCCE method does not exhibit this limitation. Furthermore, while the inclusion of augmented confusing samples provides consistent improvements for both methods, the OSCR curve of the proposed approach remains consistently above that of ARPL + CS.

C. OOD Detection

To examine whether the improvements shown in the previous OSR experiments are agnostic to the choice of the unknown rejection algorithm, we conducted additional experiments on OOD detection. The goal here is similar to OSR, with the distinction that instead of differentiating between known and unknown classes from the same dataset, we treat all classes from an in-distribution (ID) dataset as known, while other

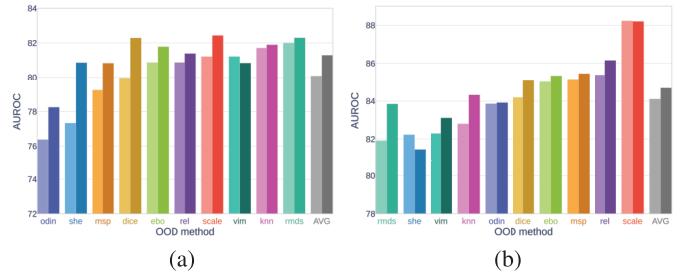


Fig. 8. Pairwise comparisons between CE without OCCE (left subbars) and with OCCE (right subbars) across different posthoc OOD methods, sorted by their baseline AUROC. (a) CIFAR-100. (b) ImageNet-200.

datasets represent OOD samples. The objective is to correctly classify ID samples while effectively detecting OOD instances.

For the ID datasets, we use CIFAR-100 and ImageNet-200, a subset of the full ImageNet-1k consisting of 200 classes with full 224×224 image dimensions. For OOD datasets, we employ CIFAR-10, TinyImageNet, and SVHN for CIFAR-100, while SSB Hard, NINCO, and iNaturalist for ImageNet-200. All experiments are conducted using a standard ResNet-18 model. We select ten competitive posthoc OOD detection methods from the literature and utilized their implementations from the OpenOOD library to ensure fairness [50]. We compare the performance of training with CE versus CE + OCCE with ($\gamma = 0.1$).

As shown in Table V, adding OCCE to the overall objective improves AUROC across all ID/OOD dataset combinations. Fig. 8 illustrates that CE + OCCE outperforms CE in nine out of ten OOD methods on CIFAR-100 and in eight out of ten methods on ImageNet-200, demonstrating consistent improvements, largely agnostic to the specific posthoc OOD method used.

VI. SENSITIVITY AND BEHAVIORAL ANALYSIS OF OCCE LOSS

A. Effect on NC

To study the effect of OCCE loss on the occurrence of NC during optimization, we measure NC using the four NC

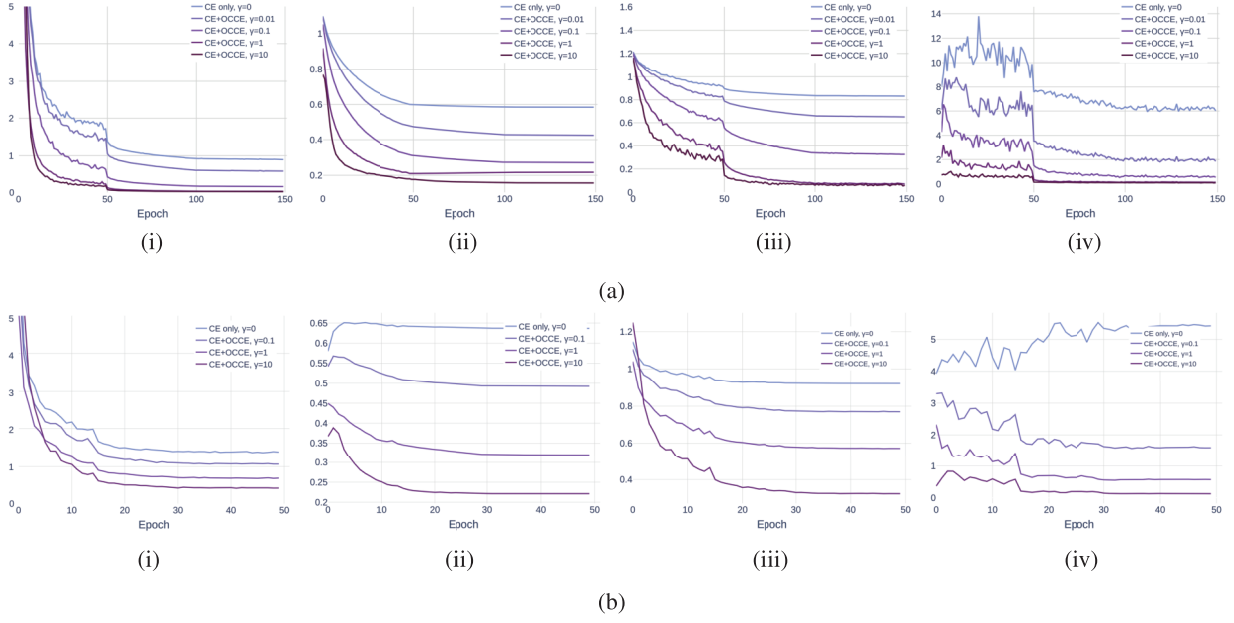


Fig. 9. Sensitivity analysis on NC during the training of ResNet18v2 and Swin-T on CIFAR-100. (a) ResNet18v2. (b) Swin-T (transfer learning). (i) $\mathcal{NC}_1$. (ii) $\mathcal{NC}_2$. (iii) $\mathcal{NC}_3$. (iv) $\mathcal{NC}_4$.

TABLE V
AVERAGE AUROC ACROSS TEN POSTHOC OOD
METHODS AND THREE RUNS

ID Dataset	OOD Dataset	AUROC ($\uparrow$)	
		CE ($\gamma = 0$)	CE w/ OCCE ($\gamma = 0.1$)
CIFAR-100	CIFAR-10	77.49	78.23
	TinyImageNet	81.25	82.34
	SVHN	81.52	83.30
ImageNet200	SSB Hard	77.11	78.01
	NINCO	82.97	83.56
	iNaturalist	92.27	92.51

metrics proposed by Zhu et al. [14]. The global mean $\mathbf{h}_G$ and class means $\mathbf{h}_k$ of the last-layer features $\{\mathbf{h}_{k,i}\}$ are defined

$$\mathbf{h}_G = \frac{1}{nK} \sum_{k=1}^K \sum_{i=1}^n \mathbf{h}_{k,i}, \quad \mathbf{h}_k = \frac{1}{n} \sum_{i=1}^n \mathbf{h}_{k,i}, \quad \text{for } 1 \leq k \leq K.$$

- 1) *Within-Class and Between-Class Variabilities* ($\mathcal{NC}_1$): The within-class and between-class covariance matrices are

$$\begin{aligned} \Sigma_W &= \frac{1}{nK} \sum_{k=1}^K \sum_{i=1}^n (\mathbf{h}_{k,i} - \mathbf{h}_k)(\mathbf{h}_{k,i} - \mathbf{h}_k)^\top \\ \Sigma_B &= \frac{1}{K} \sum_{k=1}^K (\mathbf{h}_k - \mathbf{h}_G)(\mathbf{h}_k - \mathbf{h}_G)^\top. \end{aligned}$$

Then, $\mathcal{NC}_1$ is the mean trace of $\Sigma_W \Sigma_B^\dagger$, quantified by

$$\mathcal{NC}_1 = \frac{1}{K} \text{trace}(\Sigma_W \Sigma_B^\dagger).$$

- 2) *Simplex ETF Convergence* ($\mathcal{NC}_2$): The alignment of the learned classifier $\mathbf{W}$ with a Simplex ETF is assessed by

$$\mathcal{NC}_2 = \left\| \frac{\mathbf{W}\mathbf{W}^\top}{\|\mathbf{W}\mathbf{W}^\top\|_F} - \frac{1}{\sqrt{K-1}} \left(\mathbf{I}_K - \frac{1}{K} \mathbf{1}_K \mathbf{1}_K^\top \right) \right\|_F.$$

- 3) *Convergence to Self-Duality* ($\mathcal{NC}_3$): The alignment between the classifiers $\mathbf{W}$ and the centered class-means $\mathbf{H}$ to quantify self-duality is measured by

$$\mathcal{NC}_3 = \left\| \frac{\mathbf{W}\mathbf{H}}{\|\mathbf{W}\mathbf{H}\|_F} - \frac{1}{\sqrt{K-1}} \left(\mathbf{I}_K - \frac{1}{K} \mathbf{1}_K \mathbf{1}_K^\top \right) \right\|_F.$$

- 4) *Collapse of the Bias* ($\mathcal{NC}_4$): When the global mean $\mathbf{h}_G$ is nonzero, the bias $\mathbf{b}$ may compensate for this, measured as

$$\mathcal{NC}_4 = \|\mathbf{b} + \mathbf{W}\mathbf{h}_G\|^2.$$

Fig. 9 presents NC metric measurements during the standard training of ResNet18v2 and the transfer learning of Swin-T on CIFAR-100 for various γ values. Higher γ consistently results in stronger NC, indicating that OCCE amplifies this effect. These observations align with the theoretical analysis in Section III-D and empirically confirm that the OCCE objective enforces a Simplex ETF structure, as increasing γ consistently drives $\mathcal{NC}_2$ to lower values for both models.

B. Independence Deficit

OCCE encourages each class to be equidistant from all the other classes, enforcing decoupled classes with linearly independent representations. In order to show the effect on independence deficit, we follow the procedure from [9] to estimate the intrinsic dimension of the penultimate classification layer required to reconstruct the original classification accuracy. This involves computing the covariance matrix of the representations in that layer, extracting its principal components, and projecting onto subspaces spanned by the top-K eigenvectors.

Fig. 10(a) shows that with standard CE loss, 80%, 90%, and 99% of the original classification accuracy can be reconstructed using only 22, 37, and 64 dimensions for a ResNet,

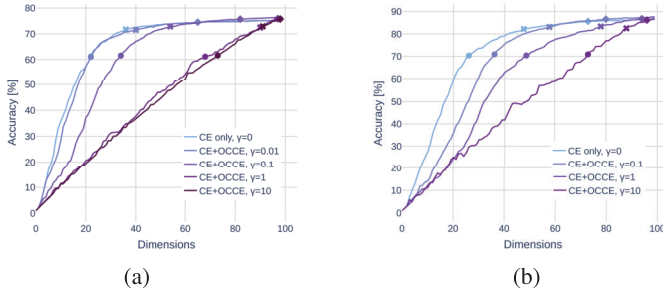


Fig. 10. Sensitivity analysis on independence deficit for converged ResNet18v2 and Swin-T. The reconstructed accuracy is plotted based on the top-K eigenvectors of the activation layer, with $\circ$, $\times$, and $\diamond$ indicating the points where 80%, 95%, and 99% of the original accuracy is reconstructed, respectively. (a) ResNet18v2. (b) Swin-T.

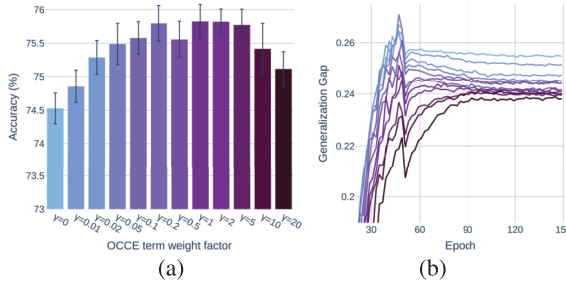


Fig. 11. Sensitivity analysis on validation accuracy and the generalization gap between train and test accuracies (ten runs). The configuration used for these experiments is a ResNet18v2 trained on CIFAR-100. (a) Accuracy. (b) Generalization gap.

respectively, from the original 100-D activation layer on CIFAR-100. However, when using OCCE and increasing the value of γ , the reconstructed accuracy shows a more linear relationship with the dimensionality of the subspace. This suggests that OCCE decouples the classes and enforces linear independence in the activation layer of each class. With $\gamma = 10$, reconstructing 80%, 90%, and 99% of the accuracy requires 73, 89, and 98 dimensions for a ResNet18, respectively. The same behavior is observed for a Swin-Tiny Vision Transformer. Intuitively, OCCE encourages each class c_i to be identified by the corresponding activation z_i , rather than permitting predictions for c_i through linear combinations of other activations $z_j, j \neq i$.

C. Generalization and Sensitivity Analysis

Our evaluation demonstrates that OCCE effectively controls the degree of NC and independence deficit in our models. But is increasing NC or reducing the independence deficit always beneficial for generalization? This process forces the network to fully decouple class vectors, potentially disregarding meaningful similarities, which may not be ideal. As shown in Fig. 11(b), increasing γ consistently narrows the generalization gap between train and test accuracies, indicating a strong correlation between higher OCCE contribution and improved generalization. However, when γ becomes too large, it limits the achievable training accuracy, as the optimization's difficulty increases with OCCE's influence, resulting in lower validation accuracy, as shown in Fig. 11(b). The network

TABLE VI
NC SENSITIVITY ANALYSIS. RESULTS ARE AVERAGES OVER TEN SEEDS IN CIFAR-100 WITH RESNET18V2

Config	CE	$\gamma = 0.1$	$\gamma = 1$	$\gamma = 10$	Simplex ETF
Accuracy ($\uparrow$)	74.51	75.54	75.78	75.40	74.42

converges on solutions that sacrifice accuracy in favor of better alignment of complementary classes.

In Table VI, we report the accuracy of CIFAR-100 when training a ResNet18v2 with CE, with CE + OCCE using different values of γ , and with a predefined and fixed Simplex ETF classifier trained with CE. While a Simplex ETF represents the theoretically optimal solution, predefined the structure of the classifier in this manner does not lead to improvements over CE. However, it provides computational benefits, as we do not need to train the last layer of the network. The parameter γ allows us to control the influence of OCCE and, consequently, the degree of NC induced. As a result, there exists an optimal range for γ , which depends on the dataset's complexity and the model's architecture.

In our experiments, we fixed $\gamma = 1$ unless otherwise specified, to keep the setup as simple as possible. However, since γ plays a critical role in the effectiveness of our approach, careful tuning via grid search based on the validation CE loss is a straightforward yet effective strategy for selecting an appropriate value. Empirically, we propose restricting the grid search range between 0.1 and 1 to improve search efficiency.

VII. CONCLUSION

In this article, we highlight the importance of controlling relationships between complementary classes. To this end, we introduce the concept of anticlasses as a complementary approach to modeling classification problems. By representing anticlasses using one-cold encoding, we optimize neural networks to directly learn target anticlass probability distributions through OCCE loss. Inspired by the phenomenon of NC, we focus on a uniform one-cold encoding and analyze its asymptotic convergence and gradient behavior. We prove that this approach directly enforces equal logits on the complementary classes—an assumption regularly made in the theoretical literature on CE and NC but not always observed empirically. We, then, demonstrate how class and anticlass learning can be effectively combined to improve classification in both closed-set and open-set scenarios. Finally, we show that uniform OCCE can be leveraged to induce NC and mitigate the independence deficit, while also discussing its implications for neural network generalization.

This work opens several research directions. The uniform distribution is the most unbiased prior for a target anticlass distribution when no additional knowledge about class relationships is available. However, many real-world problems involve interclass or hierarchical relationships, which Simplex ETF and uniform OCCE do not inherently capture. The application of soft, nonuniform anticlass distributions could provide a more structured and flexible class representation. For example, other priors can readily be employed as anticlass

target distributions, either given explicitly through domain knowledge or discovered implicitly from the network's own predictions via self-distillation [52]. In imbalanced settings, appropriately modifying the anticlass distributions for majority and minority classes may help mitigate the minority collapse problem. Finally, while this study derives anticlass supervision from class labels, real-world weakly supervised learning scenarios exist where only some negative labels are known. In such cases, the application of anticlasses could provide a meaningful way to enhance learning under partial supervision.

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